Your grade: 100%

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Next item →

1. What is a core reason for ReLU's popularity over sigmoid in deep networks? 1 / 1 point

- It outputs probabilities
- It avoids vanishing gradients
- It doesn't require labels
- It is smoother

Nice work

Correct: ReLU does not saturate for positive inputs, so it helps mitigate vanishing gradient issues.

2. What is the sigmoid(0)? 1 / 1 point

- 0.5
- 1
- 0
- 0.25

Nice work

Correct: The sigmoid function maps 0 to 0.5, which is its midpoint.

3. Which of the following activation functions produce non-zero outputs for negative inputs? (Select all that apply) 1 / 1 point

- [ ] ReLU
- [x] Leaky ReLU

This is one of the correct answers: Leaky ReLU gives a small negative output for negative inputs, preventing neuron death.

Nice work

- [x] ELU

Nice work

This is one of the correct answers: ELU returns smooth negative values, helping gradients flow.

- [x] Sigmoid

Nice work

This is one of the correct answers: Sigmoid produces non-zero values between 0 and 1, even for negatives.

4. Deep learning requires fewer samples to train well. 1 / 1 point

- False
- True

Nice work

Correct: A key drawback of deep learning is its data hunger, especially in supervised learning settings.

5. Which of the following describe the tanh activation function? (Select all that apply) 1 / 1 point

- [x] Output bounded between -1 and 1

Nice work

This is one of the correct answers: The output range of tanh is [-1, 1], helping control signal magnitude.

- [x] Zero-centered output

Nice work

This is one of the correct answers: Tanh is centered at zero, making it useful for data normalization.

- [x] Symmetric about origin

Nice work

This is one of the correct answers: Tanh is odd symmetric, i.e., tanh(-x) = -tanh(x).

- [ ] Always positive

6. What are major challenges with deep learning? (Select all that apply) 1 / 1 point

- [x] High computation cost

Nice work

This is one of the correct answers: Deep learning often requires GPUs or TPUs, which increases cost.

- [x] Interpretability

Nice work

This is one of the correct answers: Deep models act as black boxes, making it difficult to understand their decisions.

- [ ] Requires extensive feature engineering
- [x] Need for large training data

Nice work

This is one of the correct answers: While deep learning models are trained over large datasets, they often don't need as much labeled data now-a-days.

7. Why is the sign function rarely used in deep learning today? 1 / 1 point

- It is probabilistic
- It is not differentiable
- It uses a fixed learning rate
- It is linear

Nice work

Correct: The main reason it's avoided in deep learning is that it's not differentiable, which makes gradient-based learning (backpropagation) impossible.

8. A neuron receives inputs [2, -1] with weights [0.5, 0.7] and a bias of 1. What is the total input to the neuron before activation? 1 / 1 point

- 2.3
- -0.3
- 0.3
- 1.3

Nice work

Correct: Weighted sum = (2×0.5) + (-1×0.7) = 1 - 0.7 = 0.3. Adding bias (1) gives a total

9. You observe that your deep learning model isn't learning well from a small dataset. What is the most likely cause? 1 / 1 point

- Deep models ignore input features
- Deep models don't require data to generalize
- Small datasets can lead to overfitting in deep models
- The model hasn't been trained long enough

Nice work

Correct: Small datasets can cause deep models to memorize training data instead of learning general patterns, this is called overfitting.

10. Scenario: Your neuron's input sum is 0. What will be the output for the following activation functions? 1 / 1 point

- ReLU = 0, Sigmoid = 0.5, Tanh = 0
- ReLU = 0, Sigmoid = 0, Tanh = 0.5
- ReLU = 1, Sigmoid = 0.5, Tanh = 1
- ReLU = 0.5, Sigmoid = 0.5, Tanh = 0

Nice work

Correct: ReLU(0) = 0; Sigmoid(0) = 1 / (1 + e^0) = 0.5; Tanh(0) = 0.

11. You're comparing ReLU and Sigmoid activations. Which would you choose to reduce vanishing gradients and why? 1 / 1 point

- Sigmoid, because it always gives a non-zero output
- ReLU, because it ignores weights entirely
- Sigmoid, because it smooths gradients
- ReLU, because it doesn't saturate for positive inputs

Nice work

Correct: ReLU outputs are unbounded for positive values, allowing gradients to flow freely, which reduces vanishing gradient issues.

12. Scenario: You remove all activation functions from your deep neural network. What happens? 1 / 1 point

- It behaves like a complex non-linear model
- The model becomes interpretable
- The network reduces to a linear function, no matter how many layers
- Training becomes more efficient

Nice work

Correct: Without non-linearity, the model can only represent linear functions, even with multiple layers.

13. What is the ELU function output for x < 0? 1 / 1 point

- x
- 0
- e^x
- $\alpha(e^x - 1)$

Nice work

Correct: For x < 0, ELU = $\alpha(e^x - 1)$, adding smoothness and continuity.